Subject: Session 2.4

Year. Month. Date. ( )

Problem 17.1

$$Qx=0 \Rightarrow Q^TQx=0 \Rightarrow Ix=0 \Rightarrow x=0$$

Problem 17.2

$$A=a=(1,-1,0,0)^T$$

$$B=b-\frac{A^Tb}{A^TA}A=(0,1,-1,0)^T-\frac{(-1)}{2}(1,-1,0,0)^T$$

$$=b+\frac{1}{2}A=\left(\frac{1}{2},\frac{1}{2},-1,0\right)^T$$

$$C=c-\frac{A^Tc}{A^TA}A-\frac{B^Tc}{B^TB}B=(0,0,1,-1)^T-0+\frac{2}{3}\left(\frac{1}{2},\frac{1}{2},-1,0\right)$$

$$=\left(\frac{1}{3},\frac{1}{3},\frac{1}{3},-1\right)$$